Anand Ram S

Systems Engineer | MBSE, Controls & AI-Augmented Engineering
San Francisco Bay Area, California

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EXPERIENCE

Siemens Digital Industries Software, Fremont, California | NOV '22 - PRESENT
Senior Engineering Services Engineer

Agentic AI for Systems Engineering

- Designed and built MCP servers and copilots for Capella, System Modeling Workbench (SMW), SysML v2, and MADe that let AI agents read, author, and verify system architecture and safety analysis models — reducing manual model-authoring and linking time by more than 80%
- Developed a multi-agent framework in Simcenter AI Agents for a customer PoC that automates controls verification and validation on an Amesim and Simulink co-simulation, from test design and harness creation through simulation, reporting, and debugging — cutting manual testing time by an estimated 50%
- Developing an agentic safety-by-design workflow that generates MADe failure diagrams from SysML v2 architecture models, with AI output constrained to MADe's failure taxonomy; INCOSE journal paper in-progress
- Exploring a knowledge-driven FMEA authoring pipeline that combines Amesim model data with historical failure data mined through GraphRAG to generate failure diagrams in MADe

Solution Delivery & Digital Thread

- Technical solution architect for DENSO Corporation's global Model-Based Design (MBD) digital thread transformation, enabling the integration of requirements management, system architecture, and simulation tools (Polarion ALM, Simcenter System Architect, Rhapsody, Simcenter Amesim)
- Lead requirements engineer on an aerospace UAV prototype program using Harmony MBE in IBM Rhapsody: L0 to L2 architecture, requirements traceability, and function ID linkage to controls and simulation teams

Siemens Digital Industries Software, Livonia, Michigan | FEB '18 - NOV '22
Engineering Services Engineer

Ford Bronco AWD System, Technical Lead (Testing)

- Designed, developed, and implemented All-Wheel-Drive (AWD) controls algorithms from a clean sheet, covering software architecture, code generation, integration, and MIL/SIL testing, from concept to production using the System V-Cycle
- Defined vehicle-to-component requirements with systems engineers, suppliers, and calibrators; led client operations for agile delivery

Enterprise Transformation & Powertrain Controls

- Led Model-Based Systems Engineering (MBSE) process assessments and digital thread roadmaps for US and Japanese automotive clients
- Led multi-attribute powertrain controls optimization, balancing NVH, efficiency, and drivability trade-offs for HMC's NVH team

Ashok Leyland Ltd., Chennai, India | AUG '14 - JUNE '16
Deputy Manager, Technical Service & Strategy Division

- Resolved recurring diesel engine and synchromesh transmission failures through structured root-cause analysis with product development and reliability teams
- Spearheaded service strategy & analytics initiatives, developing data-driven models to improve workshop performance across 500+ dealerships nationwide

Graduate Engineering Trainee, New Product Development Division

- Applied P-diagram, QFD, DFMEA, FEA, and Lean Six Sigma to resolve structural issues in Class 6 and 7 truck load bodies

EDUCATION

University of Michigan, Ann Arbor, MI
Master of Engineering | SEP '16 - DEC '17
Integrative Systems + Design (ISD), Global Automotive & Manufacturing Engineering
CGPA 3.7/4.0

Anna University, Chennai, India
Bachelor of Engineering | APR '10 - APR '14
Mechanical Engineering
CGPA 9.1/10.0

SKILLS

AI & Integration

MCP servers, Multi-agent Workflows
Context Engineering for AI Agents
Claude Code, Goose / Simcenter AI Agents
Python, REST / OpenAPI, SOAP
Metamodels (ARCADIA, SysML v2)

Software & Tools

IBM Rhapsody, Polarion, Teamcenter
Simcenter System Architect, Amesim
Capella / SMW, MADe
MATLAB / Simulink / Stateflow

Methods & Practices

Requirements & Traceability
System Architecture (SysML)
Safety Analysis (FHA, FMEA, FTA)
MBD, MIL/SIL/HIL, ASPICE
Agile/Scrum, Lean Six Sigma
Cross-Cultural Program Leadership

CERTIFICATIONS

Google Project Management Specialization
Lean Six Sigma Green Belt Training Certification

AWARDS & LEADERSHIP

Ford Recognition Award | MAR '21
Ford Bronco driveline controls, Ford Motor Company, Michigan

Vice President | SEP '16 - DEC '17
Graduate Student Advisory Committee (GSAC), Integrative Systems + Design, University of Michigan, Ann Arbor

Vice President | JUNE '13 - MAY '14
Society of Automotive Engineers (SAE) College Club, Anna University, Chennai

University Gold Medalist | Class of 2014
Rank 10 (among 7124 candidates), Bachelor of Engineering, Mechanical Engineering, Anna University, Chennai, India